

- Among the 100-stock portfolios, the standard deviation was 1.11 percent—18.3 percent best, 10 percent worst.
- Among the 50-stock portfolios, the standard deviation was 1.54 percent—19.1 percent best, 8.6 percent worst.
- Among the 15-stock portfolios, the standard deviation was 2.78 percent—26.6 percent best, 6.7 percent worst.

From all this, one key finding emerged: When we reduced the number of stocks in a portfolio, we began to increase the probability of generating returns that were higher than the market's rate of return. But, not surprisingly, at the same time we also increased the probability of generating lower returns.

To reinforce the first conclusion, we found some remarkable statistics when we sorted the data:

- Out of 3,000 250-stock portfolios, 63 beat the market.
- Out of 3,000 100-stock portfolios, 337 beat the market.
- Out of 3,000 50-stock portfolios, 549 beat the market.
- Out of 3,000 15-stock portfolios, 808 beat the market.

With a 250-stock portfolio, you have a one-in-50 chance of beating the market. With a 15-stock portfolio your chances increase dramatically, to one in four.

Another important consideration: In this study, I did not factor in the effect of trading expenses. Obviously, where the portfolio turnover ratio is high, so are the costs. These costs work to lower the return for investors.

As to the second conclusion, it simply reinforces the critical importance of intelligent stock selection. It is no coincidence that the superinvestors of Buffettville are also superior stock pickers. If you run a focus portfolio and do not have good stock-picking skills, the underperformance could be striking. However, if you develop the skill set to pick the right companies, then outsized returns can be achieved by focusing your portfolio on your best ideas.

When we isolated our 3,000 focus portfolios in the statistical laboratory 15 years ago, it was a simple and straightforward exercise. Since then, academia has delved further into the concept of focus investing, examining the behavior of different sized portfolios and studying the performance returns over much longer time periods. Most notably, K. J. Martijn Cremers